

Generation Interconnection System Impact Study Report

For Queue Project Q18-1024

Lochbuie 138 kV
September 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1024. The Interconnection Customer proposes a 468.3 MW natural gas combined cycle facility interconnecting at the Lochbuie 138 kV substation in the Pennsylvania area. The request was received on July 08, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 258 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,094,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Delta Uppco Substation-North Mobile 138.0 kV
Contingency: Moser Generating Station-Owatonna 345.0 kV
Loading: 134.9% (Rating: 275 MVA)
DFAX: 18.3%
Pre-project loading: 103.7%
2. Facility: Delta Uppco-Ironwood Wind 138.0 kV
Contingency: Lochbuie-W Middleton Substation 138.0 kV
Loading: 111.2% (Rating: 243 MVA)
DFAX: 31.3%
Pre-project loading: 50.7%
3. Facility: Lochbuie-Ironwood Wind 138.0 kV
Contingency: Lochbuie-W Middleton 138.0 kV Line
Loading: 110.6% (Rating: 254 MVA)
DFAX: 31.3%
Pre-project loading: 52.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Delta Uppco Substation-North Mobile 138.0 kV	Rebuild line with higher-capacity conductor	\$1,694,000	90% (\$1,528,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Delta Uppco-Ironwood Wind 138.0 kV	Rebuild line with higher-capacity conductor	\$2,915,000	96% (\$2,807,000)
Lochbuie-Ironwood Wind 138.0 kV	Rebuild line with higher-capacity conductor	\$2,866,000	96% (\$2,759,000)
	Total allocated to project		\$7,094,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.